

Radiative decay rates of localized plasmon modes

In these notes, we compare the radiative decay rate obtained through two different methods: **Method 1**: By calculating the ratio of the power to the stored energy. **Method 2**: By calculating the complex frequency, $\omega + i\gamma$ which corresponds to the pole of the reflection coefficient.

We outline here the structure of this. In **Appendix 1**, we calculate the radiative decay rate through **Method 1**. First, we calculate the effective dipole moment of the plasmon mode and subsequently use the Larmor formula to compute the radiated power, P . We then calculate the total electromagnetic energy (which just corresponds to the total electric energy for small spheres). In **Appendix 2**, we calculate the exact radiative decay rate by finding the complex frequency which gives the pole of the reflection coefficient (we do this for small spheres and for high quality factors so that we may may comparisons with **Method 1**).

Appendix 3 and **Appendix 4** correspond (respectively) to **Method 1** and **Method 2** applied to the cylindrical system.

Notational point

In the below, at times we use the notation ε_m or n_m . These refer to permittivities and refractive indices of metals. The subscript m has no relation to any multipole moment.

Appendix 1: Method 1

Some of the notation in this section may be found in the papers Electroluminescence efficiency enhancement using metal nanoparticles (<https://pubs.aip.org/aip/apl/article/93/2/021120/336422>) (henceforth referred to as **Ref. 1** and Impact of high-order surface plasmon modes of metal nanoparticles on enhancement of optical emission (<https://pubs.aip.org/aip/apl/article/95/17/171103/321136>), henceforth referred to as **Ref. 2**, both by **Khurgin**.

If one has a uniform electric field everywhere in a polarizable sphere, one may obtain the polarization density everywhere in the sphere via: $\mathbf{P} = \mathbf{D} - \varepsilon_0 \mathbf{E} = \varepsilon_0 [\varepsilon - 1] \mathbf{E}$, which means that the total dipole moment will be given by:

$$\mathbf{p} = \varepsilon_0(\varepsilon_m - 1) \left[(4/3)\pi a^3 \right] \frac{3\varepsilon_d}{\varepsilon_m + 2\varepsilon_d} E_0 e_z = \varepsilon_0(\varepsilon_m - 1) (2\pi a^3/3) E_{max} e_z,$$

where we used expressions for the electric field from **Equation 1** of **Ref. 1**.

Note that, for $\varepsilon_d = 1$ (corresponding to a plasmon resonance at $\varepsilon_m = -2$), we have that $\mathbf{p} = -2\varepsilon_0\pi a^3 E_{max} e_z$.

The Poynting vector associated with point dipole radiation is given by:

$$\mathbf{S} = \frac{\mu_0 |\mathbf{p}|^2 \omega_0^4}{32\pi^2 c} \frac{\sin^2(\theta)}{r^2} e_r \rightarrow P = \frac{\mu_0 |\mathbf{p}|^2 \omega_0^4}{16\pi c} \int_0^\pi \sin^3(\theta) d\theta$$

We remind the reader that the easiest way of doing this integral is:

$$\sin^3(\theta) = \frac{i}{8} \left[e^{3i\theta} - e^{-3i\theta} - 3e^{i\theta} + 3e^{-i\theta} \right] \rightarrow \int_0^\pi \sin^3(\theta) d\theta = \frac{i}{8} \left[-\frac{4}{3i} + \frac{12}{i} \right] = \frac{4}{3}$$

Therefore, we get for the power, P , (substituting in $\varepsilon_m = 2\varepsilon_d$ and keeping ε_d arbitrary for now):

$$P = \frac{\mu_0 |\mathbf{p}|^2 \omega_0^4}{12\pi c} = \frac{\mu_0 \varepsilon_0^2 \omega_0^4}{12\pi c} \left[2\varepsilon_d + 1 \right]^2 (2\pi a^3/3)^2 E_{max}^2 = \frac{\mu_0 \varepsilon_0^2 \omega_0^4}{12\pi c} (2\pi a^3)^2 E_{max}^2$$

And the energy in the mode is equal to (using **Equations 3 & 4** from **Ref. 2** and setting $\varepsilon_d = 1$):

$$U = \frac{1}{4} \varepsilon_0 \varepsilon_d E_{max}^2 V_{eff} = \frac{1}{4} \varepsilon_0 (2\pi a^3) E_{max}^2$$

Therefore, the decay rate (where γ denotes decay of the field and 2γ decay of the intensity/energy), from our **Method 1** is given by:

$$-2\gamma = P(\omega_0)/U(\omega_0) = \frac{\mu_0 \varepsilon_0 \omega_0^4}{3\pi c} (2\pi a^3)$$

Note that since $\mu_0 \varepsilon_0$ has units of inverse velocity squared, γ does indeed have units of inverse time. In accordance with **Refs. 1,2**, we introduce the parameter, χ , given by:

$\chi = 2\pi a/\lambda = a\omega_0/c$, which allows us to write:

$$-2\gamma = \frac{2}{3}\omega_0(a\omega_0/c)^3 c^2 \mu_0 \varepsilon_0 = 2\omega_0 \chi^3 / 3$$

Method 2

The exact solution for the decay rate is given by $\omega = \omega_0 + i\gamma$ which solves the equation (see, for instance, [Localized plasmons in graphene-coated nanospheres](https://link.aps.org/pdf/10.1103/PhysRevB.91.125414) (<https://link.aps.org/pdf/10.1103/PhysRevB.91.125414>)):

$$h_l^1(x_2)[x_1 j_l(x_1)]' - \varepsilon_m(\omega) j_l(x_1)[x_2 h_l^1(x_2)]' = 0,$$

where $x_2 = a\omega/c$, $x_1 = \sqrt{\varepsilon_m(\omega)}a\omega/c$ are the wavevectors outside and inside the scatterer, respectively. Note that we must retain the full dispersion of the metallic nanosphere, since the electric energy computed through **Method 1** took into account dispersion (by using the **Brillouin** formula).

Our **Method 2** equation may be written as $f(\omega) + ih(\omega) = 0$, where we define $f(\omega)$ and $h(\omega)$ as follows:

$$\begin{aligned} f(\omega) &= j_l(x_2)[x_1 j_l(x_1)]' - \varepsilon_m(\omega) j_l(x_1)[x_2 j_l(x_2)]', \\ h(\omega) &= y_l(x_2)[x_1 j_l(x_1)]' - \varepsilon_m(\omega) j_l(x_1)[x_2 y_l(x_2)]' \end{aligned}$$

We use expansions of the spherical bessel functions (for small arguments) from <https://hackmd.io/@aligho/ByCdZ6A3ex> (<https://hackmd.io/@aligho/ByCdZ6A3ex>) in order to write:

$$j_1(x_1) \approx \frac{x_1}{3}, [x_1 j_l(x_1)]' \approx \frac{2x_1}{3}, [x_2 j_l(x_2)]' \approx \frac{2x_2}{3},$$

$$y_l(x_2) \approx -\frac{1}{x_2^2}, [x_2 y_l(x_2)]' \approx \frac{1}{x_2^2}$$

Which allows us to write:

$$f(\omega) \approx \frac{x_2}{3} \frac{2x_1}{3} - \varepsilon_m(\omega) \frac{x_1}{3} \frac{2x_2}{3} = \frac{2}{9} (a\omega/c)^2 n_m(\omega) (1 - \varepsilon_m(\omega))$$

$$h(\omega) \approx -\frac{1}{x_2^2} \frac{2x_1}{3} - \varepsilon_m(\omega) \frac{x_1}{3} \frac{1}{x_2^2} = -\frac{n_m(\omega)}{3a\omega/c} (2 + \varepsilon_m(\omega))$$

Therefore, we have that our **Method 2** condition may be written as, for small nanospheres:

$$\frac{n_m(\omega)}{3a\omega/c} \left[\frac{2}{3} (a\omega/c)^3 [1 - \varepsilon_m(\omega)] - i [2 + \varepsilon_m(\omega)] \right] = 0$$

For the above to be zero, we obviously need the term in brackets to be zero. We therefore expand the term in brackets up to first order in the imaginary part of the frequency:

$$\frac{2}{3} \frac{a^3}{c^3} [\omega_0^3 + 3i\omega_0^2\gamma] [3 - i\varepsilon'_m(\omega_0)\gamma] + \varepsilon'_m(\omega_0)\gamma = 0$$

Separating this condition into its real and imaginary parts then gives us:

$$2(a\omega_0/c)^3 + \varepsilon'_m(\omega_0)\gamma = 0 \rightarrow \gamma = -2\chi^3/\varepsilon'_m(\omega_0) = -\chi^3\omega_0/3$$

$$(2/3)(a^3/c^3) [-\varepsilon'_m(\omega_0)\omega_0^3 + 9\omega_0^2] \chi^3\omega_0/3 = 2\chi^6/3,$$

where we used the following expression for the derivative of the permittivity:

$$\varepsilon'_m(\omega_0) = \frac{2\omega_p^2}{\omega_0^3} = \frac{6}{\omega_0},$$

and furthermore used that the dipole plasmon frequency, $\omega_0 = \omega_p/\sqrt{3}$. Note that while the above approach did not nullify both the real and imaginary parts, it did do so up to order $\chi^3 = (a\omega_0/c)^3$. The remaining term is of order $\chi^6 = (a\omega_0/c)^6$.

Appendix 3: The cylindrical case (Method 1)

In the cylindrical case, in the quasistatic limit, (see <https://hackmd.io/@aligho/rk6siaGTeg> (<https://hackmd.io/@aligho/rk6siaGTeg>), which has the notation we use below), we have that the electric field, for a multipole, m , is given by:

$$\begin{aligned}\mathbf{E}_m(\rho < a, \theta) &= b_m \left[m\rho^{m-1} e^{im\theta} e_\rho + im\rho^{m-1} e^{im\theta} e_\theta \right] \rightarrow \\ \mathbf{E}_1(\rho < a, \theta) &= b_1 e^{im\theta} (e_\rho + ie_\theta) = b_1 (e_x + ie_y) \\ \mathbf{E}_1(\rho > a, \theta) &= b_1 a^2 \left[-\frac{1}{\rho^2} e_\rho + \frac{i}{\rho^2} e_\theta \right] e^{i\theta}\end{aligned}$$

Note that we need to have $\varepsilon_m(\omega) = -1$ (if the outside of the cylinder is vacuum) in order to maintain continuity of the displacement field's radial component. We calculate the energy stored in the electric field per unit length:

$$U = \frac{1}{4} \varepsilon_0 |b_1|^2 a^4 \int_a^\infty \frac{1}{\rho^4} 4\pi \rho d\rho + \frac{3}{2} \varepsilon_0 |b_1|^2 (\pi a^2) = 2\varepsilon_0 |b_1|^2 a^2 \pi$$

Note that, in the above, we implicitly used the Brillouin formula for the energy stored in the cylinder (i.e. we used $\partial_\omega(\omega\varepsilon_m(\omega))$ instead of $\varepsilon_m(\omega)$). In addition, at the surface plasmon frequency, $\omega_p^2 = 2\omega_0^2$ with ω_0 the surface plasmon frequency. In addition, $\partial_\omega(\omega\varepsilon_m(\omega))|_{\omega_0} = 1 + \omega_p^2/\omega_0^2 = 3$.

We now need to find the Poynting vector in order to find the radiated power. This obviously cannot be done quasistatically, so we need to solve for the exact electromagnetic fields.

The exact magnetic field is given by $\mathbf{H} = H e_z$, $H = \beta e^{im\theta} J_m(n_m \omega/c)$ inside the cylinder, which means that the electric field (for the dipole mode) will be given by

$$\mathbf{E} = \frac{i\beta}{\omega\epsilon_0\epsilon_m(\omega)} \left[\frac{i}{\rho} J_1(n_m\omega\rho/c)e_\rho - \frac{n_m\omega}{c} J_1'(n_m\omega/c)e_\theta \right] e^{i\theta} \approx$$

$$\frac{i\beta}{2\omega\epsilon_0\epsilon_m(\omega)} \left[ie_\rho - e_\theta \right] \frac{n_m\omega}{c} e^{i\theta} = -\beta \frac{e_\rho + ie_\theta}{2c\epsilon_0 n_m(\omega)} e^{i\theta},$$

where we used expansions of the Bessel function for small arguments. Note that from this expression, we see immediately the relationship between β and b_1 , which we defined above:

$$\beta = -2c\epsilon_0 n_m(\omega) b_1$$

To calculate the Poynting vector, $\frac{1}{2}\Re(\mathbf{E} \times \mathbf{H}^*) = \frac{1}{2}\Re(\mathbf{E}_\theta \times \mathbf{H}_z^*)$, we need to find the fields outside the cylinder asymptotically far away.

Outside the cylinder, we have $H_z = \alpha e^{i\theta} H_1^{(1)}(\rho\omega/c)$, where the coefficient α is given by:

$$\alpha = \beta J_1(n_m\omega a/c)/H_1(\omega a/c) \rightarrow |\alpha| = |\beta| \frac{|J_1(n_m\omega a/c)|}{\sqrt{|J_1(n_m\omega a/c)|^2 + |Y_1(n_m\omega a/c)|^2}}$$

$$\rightarrow |\alpha| \approx |\beta| \pi(\omega a/c)^2 |n_m|/4$$

Furthermore, we have that the azimuthal component of the electric is given by:

$$E_\theta = -i\alpha e^{i\theta} H_1^{(1)}(\rho\omega/c)'/(c\epsilon_0),$$

which means that the Poynting vector, $\mathbf{S}$ and the radiated power, P will be given by:

$$\frac{e_z}{2} i|\alpha|^2 H_1^{(1)}(\rho\omega/c)' H_1^{(1)}(\rho\omega/c)^*/c\epsilon_0 \approx \frac{|\alpha|^2}{\pi\rho\omega\epsilon_0} e_z \rightarrow P = 2|\alpha|^2/\omega\epsilon_0 =$$

$$(2/\omega\epsilon_0) |2c\epsilon_0 b_1|^2 \pi^2 (a\omega/c)^4/16 = (a\omega/c)^4 |b_1|^2 \pi^2 (c^2\epsilon_0/\omega)/2$$

From this, we get that the radiated power per unit length divided by the energy per unit length is given by:

$$2\gamma = P/U = \frac{(a\omega/c)^4 |b_1|^2 \pi^2 (c^2/\omega)}{4|b_1|^2 a^2 \pi} = \pi\omega (a\omega/c)^2 / 4$$

Appendix 4: The cylindrical case (Method 2)

In analogy with the spherical case, we find the imaginary part of the frequency through the boundary condition:

$$J'_m(n_m ka) H_n^{(1)}(ka) - n_m J_m(n_m ka) H_n^{(1)}(ka)' = 0$$

We use the asymptotic expansions of the cylindrical Bessel functions,

$$J_1(x) \approx \frac{x}{2} \rightarrow J'_1(x) \approx \frac{1}{2},$$

$$Y_1(x) \approx -\frac{1}{\pi} \frac{2}{x}, Y'_1(x) \approx \frac{2}{\pi} \frac{1}{x^2},$$

which allows us to write the boundary condition as:

$$\left[\frac{1}{2} \frac{a\omega}{c} - \frac{2i}{\pi(a\omega/c)} \right] - \varepsilon_m(\omega)(a\omega/c) \left[\frac{1}{2} + \frac{2i}{\pi(a\omega/c)^2} \right] = 0$$

We write this equivalently as:

$$\left[1 - 4i/\pi(a\omega/c)^2 \right] - \varepsilon_m(\omega) \left[1 + 4i/\pi(a\omega/c)^2 \right] = -\frac{4i}{\pi(a\omega/c)^2} \left[1 + \varepsilon_m(\omega) \right] +$$

$$\left[1 - \varepsilon_m(\omega) \right] = 0$$

We write $\varepsilon_m(\omega) \approx \varepsilon(\omega_0) + (2\omega_p^2)(i\gamma)/\omega_0^3 = -1 + 4i\gamma/\omega_0$, which gives us:

$$\frac{16\gamma}{\pi\omega_0} + \left[a^2(\omega_0^2 + 2i\omega_0\gamma)/c^2 \right] \left[2 - 4i\gamma/\omega_0 \right]$$

We define $\tilde{\gamma} = \gamma/\omega_0$ in order to write:

$$(8/\pi)\tilde{\gamma} + (a\omega_0/c)^2 \left[1 + 2i\tilde{\gamma}\right] \left[1 - 2i\tilde{\gamma}\right] = 0 \rightarrow \gamma \approx -(\pi/8)\omega_0\chi^2,$$

where we have defined $\chi = (a\omega_0/c)$